

Balaji CRM

An internal CRM for a small Mumbai real-estate brokerage: what it is, who uses it, every screen each role reaches, and the flows that carry the work. With a master prompt for the redesign.

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PRIMARY SURFACE

Mobile PWA, 375px first

INVENTORY SOURCE

WhatsApp groups, read automatically

LIVE ROLES

Owner · Agent · Cold Caller

STACK

Next.js · FastAPI · Postgres

01 What the application actually is

Start with the business, because the software only makes sense as an answer to it. A brokerage owner runs a team of closing agents and cold callers. The work happens on phones. Inventory arrives as free-text messages in broker WhatsApp groups; leads arrive as spreadsheets and phone calls; everything else lives in someone's head.

• PROBLEM ONE

Inventory is scattered

Listings sit in hundreds of groups as unstructured text. No agent can answer “what have we got, 2BHK, Malad, under 1.5 Cr” without scrolling for an hour.

• PROBLEM TWO

The owner is blind

Nobody can say which agent showed which flat to which client, or whether a lead was ever called back. Accountability is anecdotal.

• PROBLEM THREE

The list can walk

With no access control and no audit trail, any member of staff can copy the firm's entire client book on their way to a competitor.

THE SHAPE OF THE ANSWER

Three different applications behind one login

This is the single most important thing to understand about the product. A cold caller and the owner do not see the same app with some buttons greyed out — they see genuinely different surfaces, with different navigation, built on data the server will not hand over to the other role. The role is not a UI preference. It is enforced twice, independently, in the API.

What it is not. Not a buyer-facing portal, not a native app, and not a commission or document system. It is an internal tool for one firm, designed to be operated one-handed, on a phone, in the sun, in under fifteen seconds per action.

02 Who uses it, and what each of them sees

• OWNER

Sees everything

The firm's head. Every lead, every agent, every property, the audit log, and the WhatsApp configuration. The only role that can delete, export, import, reassign, or read the raw message feed.

• AGENT

Sees their own book

Closing and field staff. Their assigned leads, the firm's shared inventory, and their showings. Cannot export, cannot reassign a lead to themselves, cannot delete a listing.

• COLD CALLER

Sees one name at a time

The calling team. A queue of assigned leads — name and number — and nothing else. No lead book, no inventory. Deliberately.

Why the cold caller sees so little

A caller who can page through the whole lead book or the firm's inventory can copy either, and neither is needed to make a call. Under a threat model whose adversary is a person who already has a legitimate account, that access is pure downside. So `contacts.browse` and `properties.read` are withheld from the role entirely — the list endpoints refuse it. Hiding the navigation is a consequence, not the enforcement.

Capability matrix

CAPABILITY	OWNER	AGENT	COLD CALLER
<code>contacts.browse</code>	✓	✓	—
<code>properties.read</code>	✓	✓	—
<code>contacts.create / edit</code>	✓	✓	✓
<code>calls.log</code>	✓	✓	✓

CAPABILITY	OWNER	AGENT	COLD CALLER
properties.write	✓	✓	—
properties.delete	✓	—	—
contacts.delete	✓	—	—
contacts.export	✓	—	—
contacts.reassign	✓	—	—
escalations.receive	✓	—	—
audit.read	✓	—	—
whatsapp.manage	✓	—	—

Two independent layers. The table above is *capability* checking: may this role call this endpoint at all. Separately, every read is *row-scoped*: an agent's query for contacts is rewritten to their assignments before it reaches the database. Either layer alone would prevent a leak; the database connection refuses to execute an unscoped query against a guarded table, so forgetting the scope is a crash rather than a disclosure.

03 Information architecture

Every screen sits at one flat top level — there is no nested menu anywhere in the product. What differs by role is which of these destinations exist at all, and which four earn a place on the phone's thumb bar.

ROUTE	SCREEN	WHO REACHES IT
/	Home — a different dashboard per role	All
/queue	Call queue — the next lead to ring	Cold Caller
/queue/session	Call console — one lead, full screen	Cold Caller
/my-calls	My calls — what I logged today	Cold Caller
/contacts	Leads — the lead book, filterable	Owner, Agent
/contacts/[id]	Lead detail — journey, calls, interests	Owner, Agent
/contacts/new	Add lead	Owner, Agent
/properties	Inventory — cards on phone, table on laptop	Owner, Agent
/properties/[id]	Listing detail — provenance, showings	Owner, Agent
/properties/new	Add listing — manual entry	Owner, Agent
/showings	Showings — site visits, scheduled and done	Owner, Agent
/escalations	Escalations — leads flagged for the owner	Owner
/feed	Activity — firm-wide, live	Owner
/inventory-feed	Inventory feed — WhatsApp pipeline console	Owner
/import	Import leads — spreadsheet upload	Owner
/team	Team — staff and access	Owner
/team/performance	Performance — calls, showings, escalations	Owner
/audit	Audit — who did what, when	Owner
/account	Your account — password	All

What reaches the thumb bar

A floating pill holds five targets. Everything a role may open stays reachable; the bar only decides what is one tap away rather than one tap behind **More**. The split is by what is needed *away from a desk* — escalations and the lead book are urgent in a car, the activity feed and the audit log are end-of-day reading.

OWNER

On the bar

Home · Leads ·
Inventory ·
Escalations

Behind More

Showings,
Activity,
Inventory feed,
Import, Team,
Audit, Account

AGENT

On the bar

Home · Leads ·
Inventory ·
Showings

Behind More

Account

COLD CALLER

On the bar

Home · Queue ·
My calls

Behind More

Account

On a laptop the same pill becomes a fixed left rail — icon and label, same Ink surface, same order. Not a different information architecture, just the same one turned ninety degrees so the full viewport height goes to the data.

04 User flows, step by step

Four journeys carry almost all of the product's value. They are drawn here as checkpoint timelines because that is the product's own signature element — the same vertical node-and-rail shape the app uses for a lead's pipeline, a listing's showing history, and a call's escalation chain.

A • The cold caller works the queue

The highest-volume loop in the product. Everything is tuned for the time between two calls.

1 Open the queue

The list is their assigned leads only — there is no separate queue table. Leads are grouped by *why* each surfaced, so the order is legible rather than taken on faith.

`/queue`

2 Start the session

One lead fills the screen: name, number, a tap-to-dial. No budget, no inventory, no other leads visible.

`/queue/session`

3 Log the outcome

Six tap-to-select outcomes — `connected`, `not_reachable`, `not_interested`, `interested`, `callback_requested`, `wrong_number` — plus a temperature of hot, warm or cold, and a free-text remark. Chips rather than dropdowns, because typing is the enemy.

4 The system reacts

`interested` or `callback_requested` creates a follow-up task automatically — the caller never has to remember to. The lead's stage advances, and the call is written to the activity feed the owner is watching.

✓ Escalate, or move on

Flagging for the owner pushes a notification to every owner device and places the lead in the escalation inbox. Otherwise the next lead is already loaded.

`/my-calls` records what they did today

B • The agent takes a lead to a site visit

1 Open the lead

The journey timeline shows every stage the lead has passed through, every call logged against it, and by whom.

`/contacts/[id]`

2 Match inventory

The lead's budget and location preferences are matched against shared inventory, so the agent is offered flats rather than searching for them.

3 Record interest

A property interest links lead to listing and moves through `inquired` → `site_visit_scheduled` → `site_visit_done` → `negotiating`. This link is what makes the owner's core question answerable.

4 Log the showing

Done from the phone, at the flat. Who showed what, to whom, when — the record the owner has never had before.

`/showings`

✓ Advance or close

The lead moves toward `negotiating` and `closed`, or to `lost`. Either way the journey is intact and attributable.

C • Inventory arrives by itself

No person is in this loop. It is the product's largest single piece of leverage, and section 05 covers the machinery.

1 The owner links WhatsApp

A QR appears in the CRM, produced by the gateway, scanned with the phone. The account's groups then list themselves.

`/inventory-feed`

2 The owner picks groups

Out of hundreds, a handful carry inventory. Only selected groups are read — the switch is in the product, and the ingest endpoint drops anything from a group not on the list.

3

Messages are stored verbatim

Raw text first, interpretation second. That ordering is what lets a prompt improvement be replayed over history instead of the history being lost.

4

A model reads each message

It decides whether the text is inventory at all, then pulls out location, building, configuration, price and area — for *every* flat in the message, since one post routinely carries nine.

✓

Listings merge, sightings accumulate

A flat reposted by four brokers becomes one inventory row and four recorded sightings. Merging never discards who posted what, where, or when.

`/properties`

D • The owner checks the firm

1

Escalations first

Leads a caller flagged. This is the only inbox in the product and it is deliberately the owner's landing point when something is waiting.

`/escalations`

2

Activity feed

Firm-wide and live: calls logged, showings recorded, stages moved, listings added.

`/feed`

3

Performance

Per member of staff: calls made, showings done, escalations raised.

`/team/performance`

✓

Audit when it matters

Every privileged action is recorded — exports, reassignments, deletions, WhatsApp configuration changes. This is the layer that makes the data-leak problem tractable rather than merely discouraged.

`/audit`

05 The inventory pipeline

Worth its own section because it is the part of the product that has no equivalent elsewhere, and because its failure modes are invisible by default — a pipeline that silently stops looks exactly like a quiet week in the groups.

STAGE ONE

The gateway

A small Node process holding a linked WhatsApp device. It reads selected groups, journals every message to disk the moment it arrives, and forwards batches to the API over an HMAC-signed webhook. It holds no user credential and can read nothing else in the CRM.

Its one job is to lose nothing. On reconnect it asks the CRM where each group left off and forwards everything posted since, so a laptop that slept through a night catches up rather than losing it.

STAGE TWO

The extractor

Messages are stored before being understood. A worker claims them in batches, sends them to a language model, and applies the result. Obvious chatter is filtered out before any model is called, because paying to classify “good morning” is a bill rather than a feature.

Extraction is probabilistic, so a low-confidence listing is published *and marked* rather than either silently trusted or silently dropped.

A message's state machine

A message only moves forward, and every terminal state is distinguishable — because “we read it and it wasn't inventory” and “we read it and parsing broke” need completely different responses from the owner watching the feed.

pending

Stored, not yet sent to the model.

processing

Claimed by a worker. Claims that die are reclaimed.

extracted

Produced at least one new listing.

duplicate

Parsed fine; every listing already existed.

not_listing

Group chatter, not inventory.

failed

Errored. Retried up to three times, then left alone.

Why the retry cap. Without one, a single malformed message is re-sent to the model forever. That is a bill, not a bug report. Any message can be re-extracted by hand from the feed — including ones that “succeeded” and were read wrongly, which never reach the failed state.

What the owner watches

The inventory feed console reports facts that are otherwise indistinguishable from each other: whether the gateway is connected, whether extraction credentials exist *and* whether anything is actually running with them, the queue depth, failures in the last day, listings produced, reposts merged, and claims whose worker died mid-batch. Each of those has the same symptom — a queue that will not drain — and a different remedy.

06 The data model, in the terms the product uses

ENTITY	WHAT IT HOLDS	WHY IT EXISTS SEPARATELY
Contact	A lead: name, phone, budget, location preference, stage, temperature, owner.	The unit the whole CRM is scoped around.
Contact assignment	Which staff member holds which lead, and since when.	Assignment is history, not a column — reassignment must be visible.
Property	One flat: type, listing type, location, building, config, area, price, status.	The deduplicated truth. Four reposts are one row.
Property source	One sighting of a property in one message: who posted, which group, when, the raw text.	Makes dedup non-lossy. Merging never discards provenance.
Property interest	A lead's interest in a listing, and how far it has gone.	The join that answers “who is being shown what”.
Call log	Outcome, temperature, remark, whether it was flagged for the owner.	The cold caller's entire output, and the escalation trigger.
Activity	Site visits, notes, stage changes, follow-ups.	Feeds the journey timeline and the owner's live feed.
Task	A follow-up, usually created automatically from a call outcome.	So remembering is the system's job, not the caller's.
WhatsApp group / message	Which groups are read; every raw message and its extraction state.	Raw bodies stored before parsing, so history can be replayed.
Audit log	Every privileged action, with actor, resource and detail.	The answer to the data-leak problem in the brief.

The vocabularies

LEAD STAGES

new → contacted →
site_visit_scheduled → visited →
negotiating → closed | lost

CALL OUTCOMES

connected · not_reachable ·
not_interested · interested ·
callback_requested · wrong_number

LISTINGS

rent | outright · apartment · villa
· plot · commercial · available |
blocked | sold

REVIEW STATES

auto_accepted · needs_review ·
confirmed · rejected

07 The rules the interface obeys

Every screen is judged against one question: can a busy agent do this one-handed, in under fifteen seconds, with one thumb? These follow from it.

One primary action per screen

Secondary actions go behind a menu, never beside the primary one. The busiest role on the busiest day will mis-tap otherwise.

Primary actions sit low

The most common action on a screen belongs in the bottom third, where a thumb already is — not at the top, where a mouse would be.

Tap, don't type

Chips and toggles wherever the option set is known. Free text only where it genuinely must be: remarks and notes.

Status is never colour alone

Always a colour *and* a label — colourblind-safe, and the label is what actually gets read in sun glare.

Dark surfaces mean “trust this”

Ink is reserved for what the owner most needs to rely on. If everything were a dark card, the signal would be worth nothing.

Every control answers the thumb

A press scales the control 3% in 80ms and releases over 220ms. Fast in, slow out — equal timings are what make web apps feel rubbery rather than physical.

Publishing beats withholding. It runs through the whole product. A low-confidence extracted listing is shown and flagged, not hidden — an agent would rather sanity-check a flat than never see it. The flag is what keeps that honest.

A Master prompt for the redesign

Written to be pasted whole — to a designer, an agency, or an AI coding agent. Everything in it is true of the shipped product. Where it states a constraint, that constraint is enforced in the API and cannot be designed around.

You are redesigning Balaji CRM, an internal tool used by one small real-estate brokerage in Mumbai. It is not a SaaS product, it has no marketing site, and it will never have public users. Roughly a dozen people use it, most of them on a phone, most of them outdoors.

Your job is to raise the visual and interaction quality of the product without breaking the three things it exists to do, and without making it slower to operate. Read the whole brief before proposing a direction.

Context to carry in

Sections 01 to 07 of this document are the brief. The recipient needs all of it: what the product is, the three role-surfaces, every screen and who reaches it, the fixed vocabularies, and the rules the interface already obeys.

Hard constraints — the part most likely to be violated

A design that breaks one of these is not a design that needs revision. It is a design for a different product.

- 01 **A Cold Caller must never see the lead book or the inventory.** Not filtered, not partial, not “just the matching flat”. The endpoints refuse the role. Any concept showing a caller more than one client at a time is out.
- 02 **Ownership is never self-service.** No design may offer an Agent a way to reassign a lead to themselves.
- 03 **Export, delete, import and audit are Owner-only.** Do not design a bulk action into a shared surface.
- 04 **Status is never colour alone.** Always colour and a text label — colourblind-safe, and the label is what gets read in sun glare.
- 05 **One primary action per screen.** Secondary actions go behind a menu, never beside the primary one.
- 06 **Primary actions sit in the bottom third on mobile**, where a thumb already is.

- 07 **Minimise typing.** Chips and toggles wherever the option set is known; free text only for remarks and notes.
- 08 **Every core action must work at 375px, one-handed, in under fifteen seconds.** This is the test every screen is judged against.
- 09 **It is an installed PWA on iOS.** It renders under the status bar, so every top edge needs the safe-area inset and the dock needs the home-indicator inset.
- 10 **The API is on a free tier that sleeps.** Some screens wait seconds for data. Loading is a design state, not an afterthought.

What is actually wrong today

Be specific about what is being fixed. In order of leverage:

1. **The product reads as competent but joyless.** Correct hierarchy, correct spacing, very little personality — a design system applied rather than a product designed.
2. **The three role-homes are underdeveloped.** / does the least work of any screen, and it is the first screen every user opens.
3. **Empty and loading states are thin.** A new firm's first week is mostly empty screens, and they currently say very little.
4. **The inventory feed console is dense and technical,** speaking closer to the database than to the owner.
5. **Density is uniform.** A caller between two calls and an owner reviewing at 9pm get the same information rhythm.
6. **The desktop Owner dashboard should read as a command centre** and currently does not.

What to deliver, in this order

1. A stated direction — palette, type pairing, layout concept — in a paragraph, *before* any screens.
2. The three role-homes. Highest leverage, weakest today.
3. Then the call console, lead detail, inventory list and listing detail, escalation inbox.
4. Mobile at 375px first, every time. Desktop after, and not the mobile layout stretched wide.
5. Every state for every screen: loaded, loading, empty, error — and once per role where the role changes what is shown.
6. A note on any design token changed, and why.

Do not

- Do not propose a generic SaaS dashboard. The reference points are consumer apps with confident dark surfaces and tactile cards, not admin panels.

- Do not add a navigation level. The flat hierarchy is load-bearing for one-handed use.
- Do not use colour as the only carrier of meaning anywhere.
- Do not introduce a component library whose look overrides the identity.
- Do not design screens that assume fast data.
- Do not solve “it feels dead” with more animation. That was a feedback problem and is largely fixed; the remaining gap is compositional, not kinetic.

HOW TO USE IT

Agree the direction before the work is spent

With an AI coding agent — paste the brief, then name the single screen you want built and add: *“mobile at 375px first; show me the empty and loading states too.”* The hard constraints are the part that stops a plausible design from being unbuildable.

With a human designer — send this document and ask for the stated direction before any screens.

Reviewing what comes back — check it against the hard constraints first, not against taste.

Balaji CRM — product & architecture reference

Owner · Agent · Cold Caller

Manager role is modelled but not yet activated